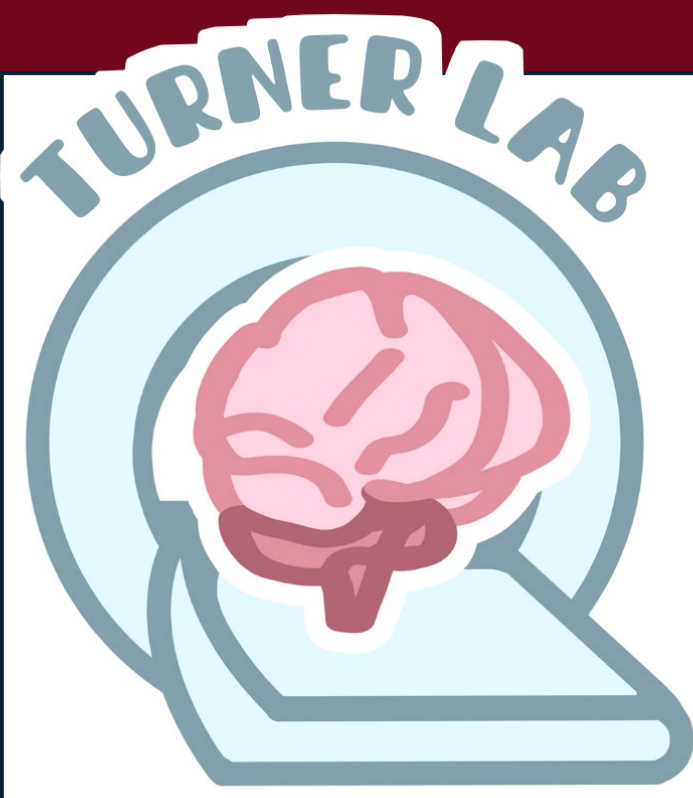




Cerebellar Transcriptomics in Schizophrenia and Bipolar Disorder Implicate Unique and Shared Genetic Mechanisms and Cortical Co-Expression Networks

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Introduction

The cerebellum shows structural and functional decreases as well as genetic expression differences in psychosis, particularly in schizophrenia.

The genetic co-expression of differentially expressed cerebellar genes may help identify both genetic processes and neural circuitry contributing to these disorders.

Methods

Differentially Expressed Genes (DEG).

- Cerebellar gene expression microarray data from cerebellar Crus I, Crus II, and Crus VI, from NCBI GEO (GSE35974) ,
- 129 individuals with schizophrenia (SZ) or bipolar disorder (BD), and healthy controls (HC), < 65 years old and with brain pH >6.0.
- DEGs defined by pairwise group differences (FDR corrected to $p < .05$).

Group (total N)	SZ (44)	BD (37)	HC (48)
Age	43.5 (20-60)	45 (20-65)	45.5 (30-60)
Sex	12 F, 32 M	18 F, 19 M	17 F, 31 M
PMI	33.3 (9-80)	35.8 (12-84)	27.6 (8-58)
Brain pH	6.44 (6-6.9)	6.45 (6-6.9)	6.49 (6-7.03)

Table 1. Demographics of the post mortem samples. (PMI = post mortem interval, in hours.)

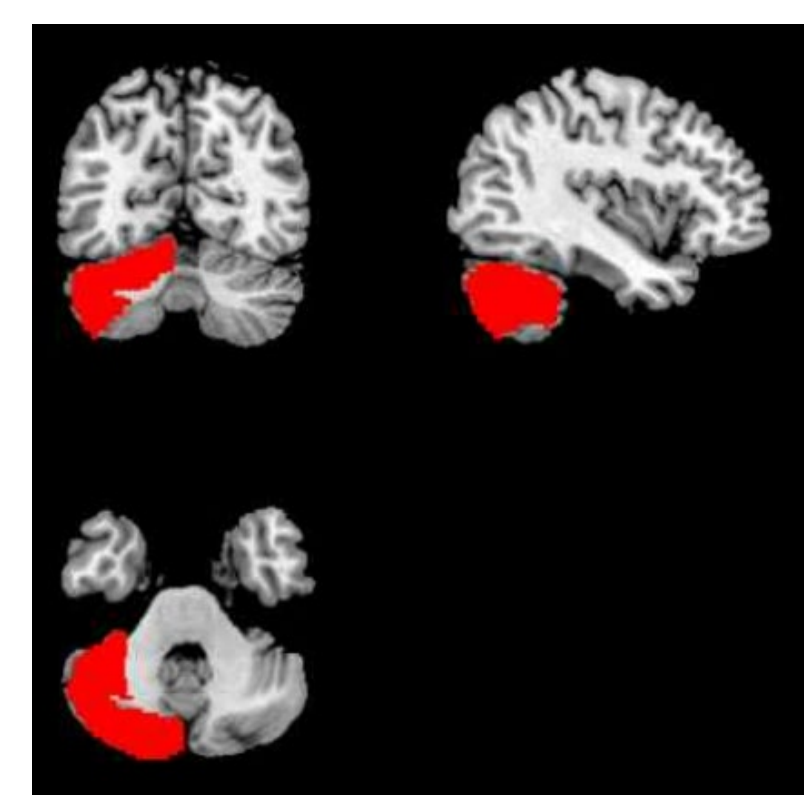


Figure 1. Cerebellar regions included in the AHBA sample.

Cortical Transcriptomic

Patterns. Using the Allen Human Brain Atlas (AHBA) left hemisphere expression data, we found average gene expression for each DEG, for the cerebellar regions and each of the 50 cortical regions from the Schaefer 100 atlas.

Results

Spatial co-expression patterns of the cerebellar DEGs from AHBA across the cortical regions were calculated using PCA; components accounting for at least 10% variance were included. The first component always captured genetic expression covariance within the cerebellum; we present here the second and third components as relevant.

SZ vs HC

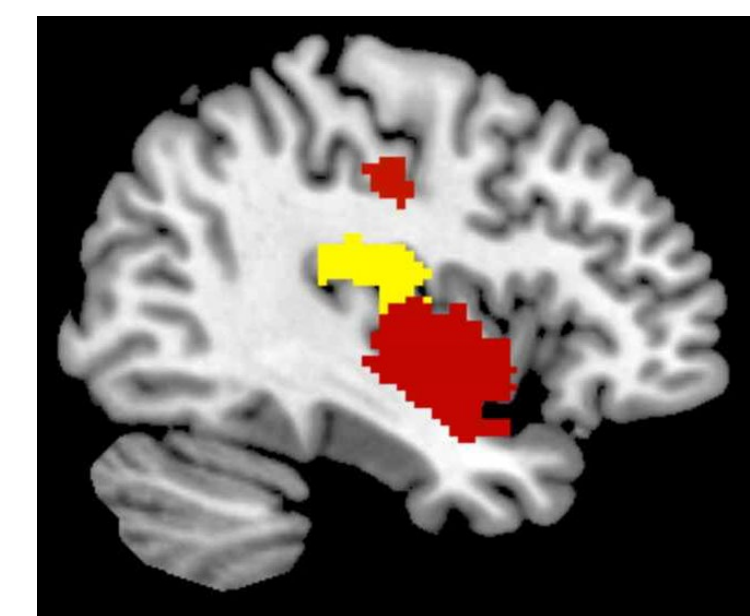


Figure 2. The DEGs from the SZ vs HC contrast showed covariation in the posterior and anterior insula, and the post central gyrus. Several genes in this component are specifically expressed in glia, astrocytes, and oligodendrocyte precursors.

BD vs HC

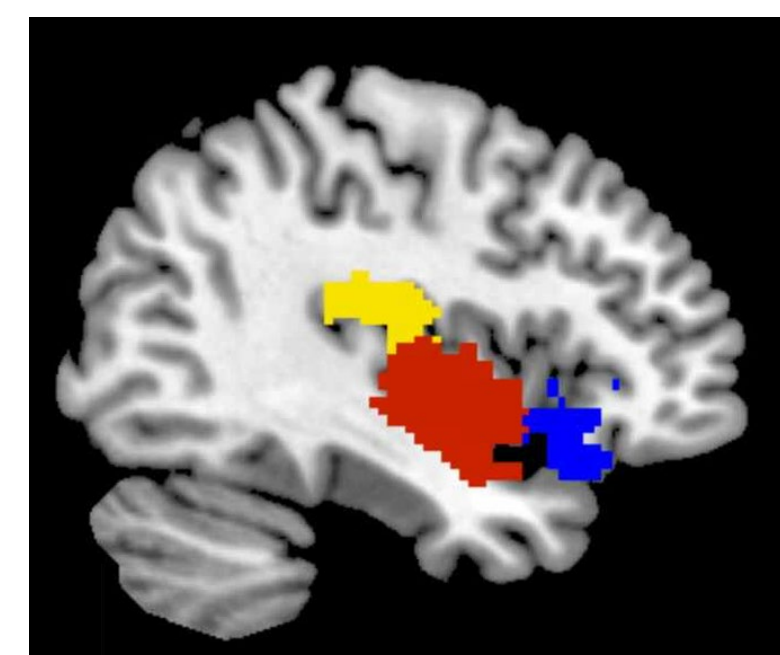


Figure 3, upper. The DEGs from the BD vs HC contrast in the second component showed covariation in the posterior and anterior insula, and a contrasting weight in the orbitofrontal cortex. Several genes in this component are specifically involved in cell maturation and differentiation but had low tissue specificity.

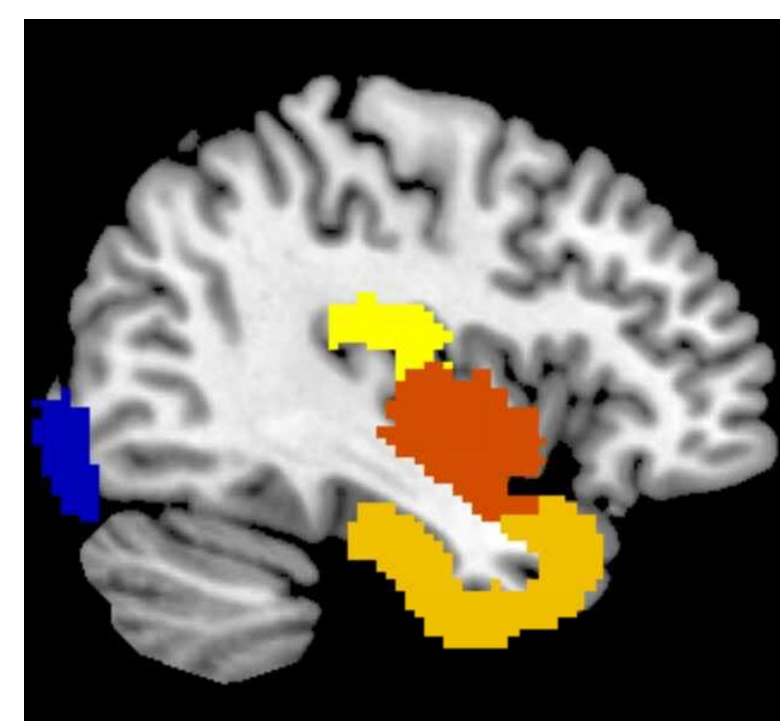


Figure 3, lower. The DEGs from the BD vs HC contrast in the third component showed covariation in the posterior and anterior insula and temporal pole, and a contrasting weight in the visual cortex and precuneus. Several genes in this component are specifically involved in cell differentiation and migration, as well as GTPase.

SZ vs BD

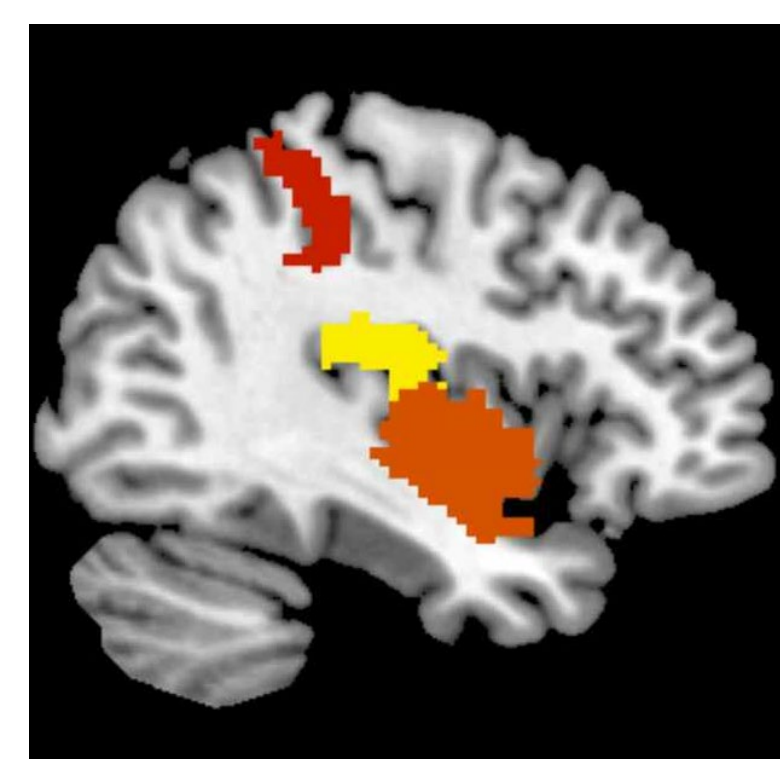
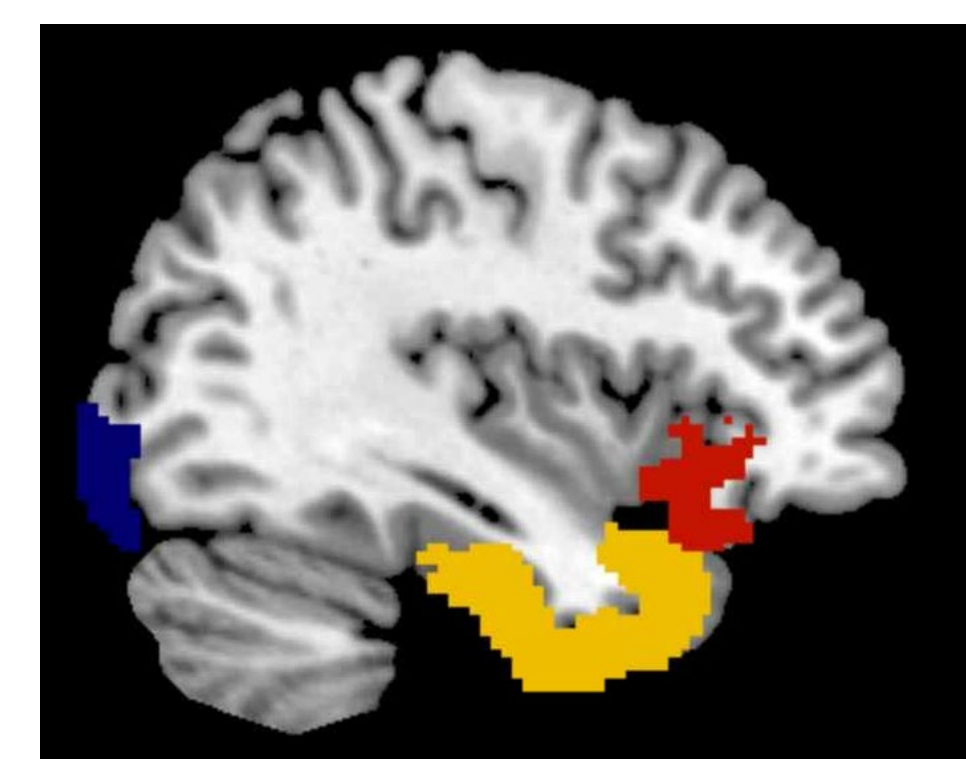


Figure 4. The DEGs from the SZ vs BD contrast in the second component (left) showed covariation in the posterior and anterior insula and precentral gyrus, with genes that were specific to brain and excitatory neurons, with contributions to cell migration and apoptosis.

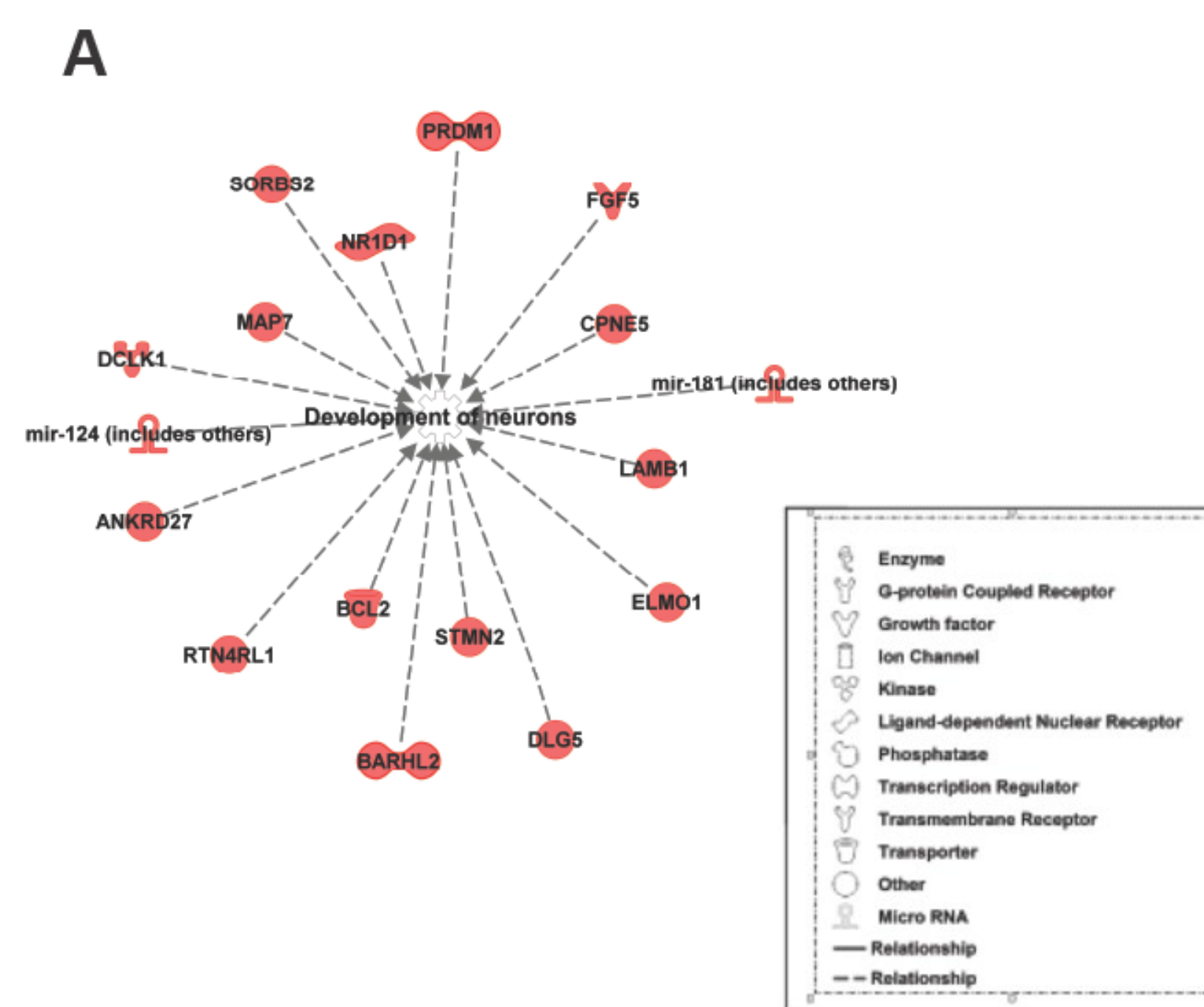


The third component (right) included the temporal pole and inferior frontal gyrus with a contrasting weight in the inferior occipital lobe. Contributing genes function largely in apoptosis, and more specifically optical neurons.

Results

Ingenuity Pathway Analysis (IPA) of DEGs are summarized below.

SZ vs HC



BD vs HC

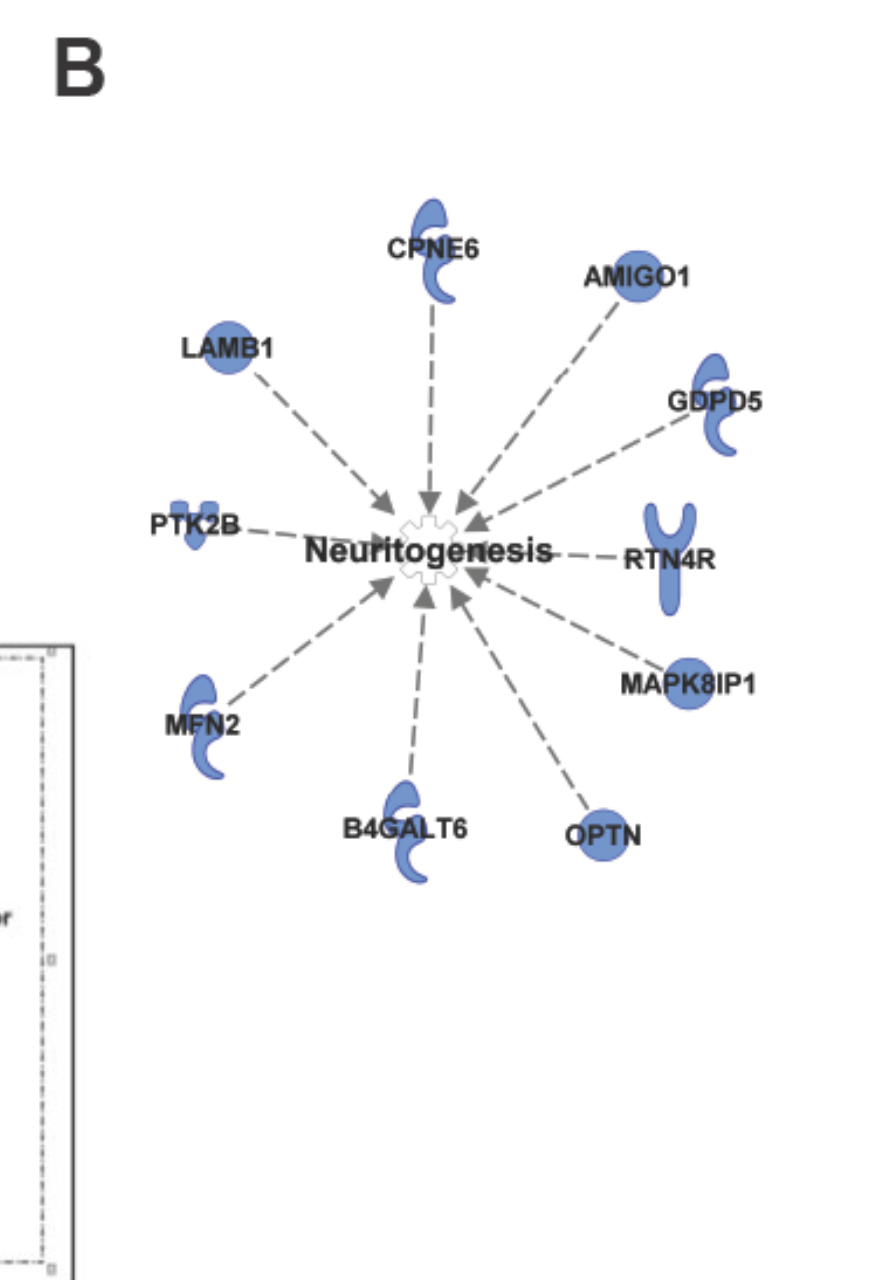
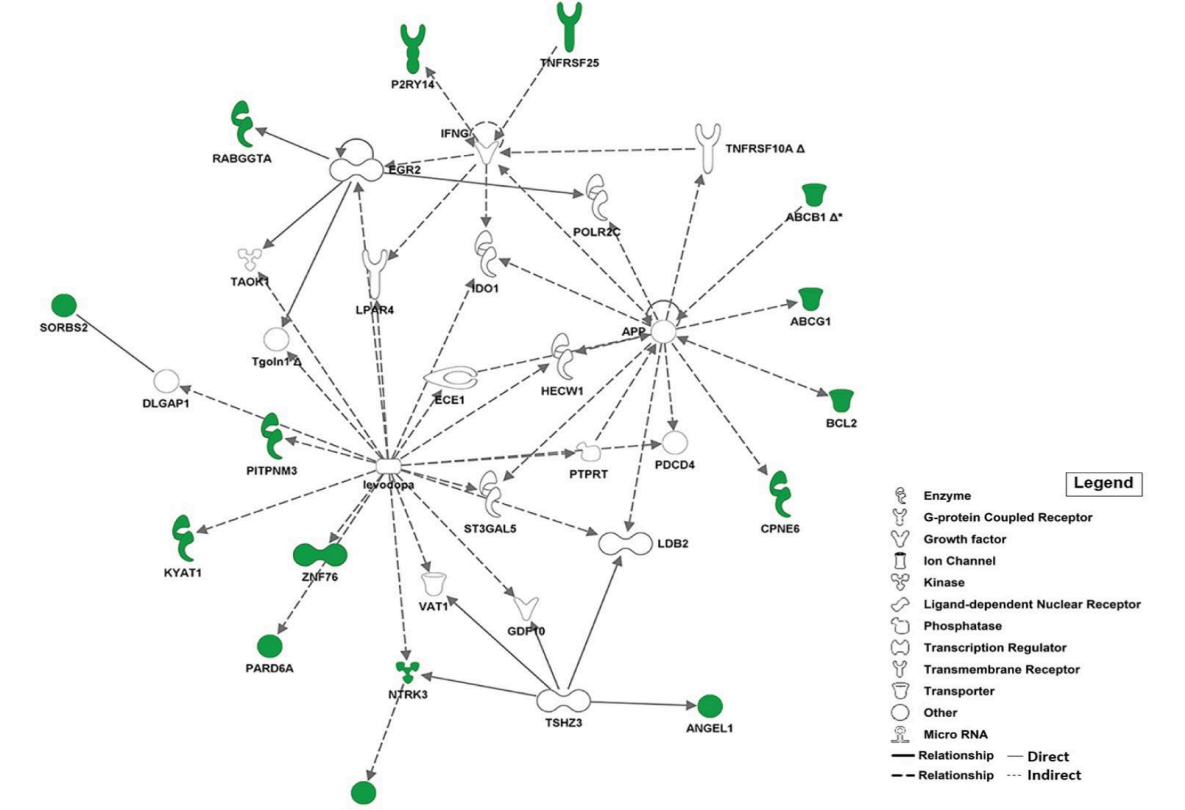


Figure 5. Development of neurons and Neuritegenesis were the top functions of the molecular networks generated by IPA analysis of DEGs in SZ vs HC and BP vs HC. The top functions from SZ (A, red) and BD (B, blue) were selected from the "Diseases and Functions" lists. The inset in the middle indicates the function of the protein molecule associated with each gene.

Figure 6. For SZ vs BD DEGs, the pathway analysis did not converge on a function, however; the associated network for this geneset is a combination of networks including "Cell Death and Survival, Inflammatory Response, and Molecular Transport".



Conclusions

- We identified differentially expressed genes (DEGs) in posterior cerebellum, across SZ, BD and HC.
- These cerebellar DEGs in the Allan Human Brain Atlas (nonpsychiatric) samples showed patterns of covarying expression that consistently implicated the anterior and poster insula.
- The DEGs networks were associated with proteins regulating cell differentiation/maturation, axonal migration, and apoptosis.

